



Elasticity of ion stuffing in chemically strengthened glass

Adama Tandia, K. Deenamma Vargheese, John C. Mauro *

Science and Technology Division, Corning Incorporated, Corning, New York 14831, USA

ARTICLE INFO

Article history:

Received 8 March 2012

Received in revised form 16 April 2012

Available online 10 May 2012

Keywords:

Ion exchange;

Elasticity;

Modeling

ABSTRACT

The chemical strengthening of glass involves the stuffing of large ions into network sites previously occupied by smaller ions. Typically this involves an exchange of Li^+ or Na^+ ions in the glass for larger K^+ ions from a molten salt bath. This swapping of ions creates compressive stress in the surface of the glass, significantly increasing the strength of the final glass product. The magnitude of this compressive stress is governed by the linear network dilation coefficient (LNDC), which defines the amount of linear strain per unit of ion substitution. However, the amount of strain attainable through ion exchange is much smaller compared to what is expected from as-melted versions of the same final glass composition. This effect, which we have termed the “network dilation anomaly,” is a result of the different local environment around the invading ion species in as-melted versus ion-exchanged glasses. A remaining question concerns the nature of the network strain due to ion stuffing. Using molecular dynamics simulations, we show that the strain induced by ion stuffing is entirely elastic. In other words, when a reverse ion exchange is performed to swap the original ions back into the glass, the initial volume of the as-melted glasses is entirely recovered. Moreover, we show that the local structural environment around the alkali ions is restored to the as-melted conditions. The elastic nature of ion stuffing demonstrates that the network dilation anomaly is not a result of plasticity, but rather a failure to achieve the full amount of expected elastic strain during ion exchange. The elasticity itself consists of both instantaneous and delayed contributions.

© 2012 Elsevier B.V. All rights reserved.

1. Introduction

The advent of ultra-thin chemically strengthened glass has revolutionized the personal electronics industry, enabling an entirely new class of touch-screen computers where glass is no longer simply a medium for visualizing information—it is now the primary interface through which the user interacts with the device. The thinness of the chemically strengthened cover glass enables high transparency, high touch sensitivity, and reduced weight for the final product. In addition, the dramatically improved strength of the glass makes it highly resistant to scratches and other damage during every day use [1].

The chemical strengthening of glass is achieved through an ion exchange process in which smaller alkali ions in the glass are replaced with larger alkali ions from a molten salt, a process originally proposed by Kistler [2] in 1962. The glass in question typically contains some concentration of a small alkali oxide, such as Li_2O or Na_2O . Upon immersion in a salt bath containing a larger alkali ion (such as K^+), the smaller alkali ions from the glass diffuse into the salt bath and are replaced with the larger ions from the salt. When these larger ions are stuffed into sites originally occupied by smaller ions in the glass, they create a compressive stress around the glass surface that

greatly improves the strength and damage resistance of the final glass product. Through this ion stuffing, the chemical strengthening process is capable of achieving a significantly higher compressive stress compared to the traditional thermal tempering approach. Ion exchange is especially well suited for strengthening of thin glass, where thermal tempering is generally not a viable option [3–6].

The compressive stress achieved through the ion exchange process is given by

$$\sigma(z) = -\frac{BE}{1-\nu} [C(z) - C_{\text{avg}}], \quad (1)$$

where E is the Young's modulus of the glass, ν is Poisson's ratio, and $C(z)$ is the local concentration of the invading alkali ion at a penetration depth z within the glass. The average concentration of the substituting cations, C_{avg} , must be subtracted from $C(z)$ so that the integral of $\sigma(z)$ over the complete thickness of the glass is zero, i.e., to satisfy the force balance condition. Thus, the integral of the negative (compressive) stress near the surface is exactly compensated by the integral of the positive (tensile) stress in the interior of the glass [7]. The parameter B in Eq. (1) is known as the linear network dilation coefficient (LNDC) [7–10] and can be considered as an analogous quantity to the thermal expansion coefficient. Whereas the thermal expansion coefficient defines the linear strain of a material

* Corresponding author. Tel.: +1 607 974 2185; fax: +1 607 974 2410.

E-mail address: mauroj@corning.com (J.C. Mauro).

per unit change in temperature, B gives the strain per unit change in alkali concentration, viz., [7]

$$B = \frac{1}{3} \left(\frac{1}{V} \frac{\partial V}{\partial C} \right) = \frac{1}{3} \frac{\partial \ln V}{\partial C}, \quad (2)$$

where V is the molar volume of the glass. The linear strain induced by ion stuffing is thus governed by the value of B , which, in turn, depends on the chemistry of the glass in question. This strain is converted into a stress via the factor of $E/(1-\nu)$ in Eq. (1).

The calculation of compressive stress may therefore seem straightforward if we can determine the value of B from Eq. (2), i.e., by measuring the difference in molar volume of as-melted (unexchanged) glasses containing different concentrations of the alkali oxides involved in the ion exchange process. However, the value of B obtained through as-melted glasses is always a factor of 2–4 higher compared with that deduced from Eq. (1) using measured values of E , ν , $\alpha(0)$, $C(0)$, and C_{avg} [5,10]. This so-called “network dilation anomaly” occurs even in the limit of short-time ion exchange experiments and therefore cannot be explained as an artifact of stress relaxation.

In a previous study [11], we addressed the detailed atomistic origin of the network dilation anomaly through molecular dynamics simulations of a series of alkali tetrasilicate glasses: $x\text{Na}_2\text{O}$ ($20-x$) K_2O 80SiO_2 (mol%). Our results revealed that the network dilation anomaly is a result of different local environments around alkali ions in as-melted versus ion-exchanged glasses. When an invading potassium ion is stuffed into a site previously occupied by a sodium ion, it is not able to adjust the glass network sufficiently to achieve the desired local environment (as would occur in an as-melted version of the same potassium-containing glass composition). The ion-exchanged glass structure thus represents a manifestation of the “forbidden glass” concept of Mauro and Loucks [12], i.e., a glass that is not attainable by cooling from a melt using any possible cooling path.

A remaining mystery concerns the nature of the strain itself during ion stuffing. In this paper, we present molecular dynamics simulations of “reverse ion exchange” to quantify the plasticity of the ion exchange process. In our reverse ion exchange simulations, the invading alkali ions are swapped with the original ions of the unexchanged glass. After this reverse exchange process is completed, we find that the original volume of the as-melted glass is fully recovered. Hence, the strain induced by ion stuffing is entirely elastic in nature. The network dilation anomaly should thus be viewed as a failure to achieve a full elastic strain, rather than a result of plasticity in the glass network, as has been suggested previously [13–17]. Owing to the nonlinear composition dependence of the elastic strain, the network dilation due to ion stuffing can be considered as an example of nonlinear elasticity [18]. Moreover, we show that the response of the glass to ion stuffing consists of both instantaneous and delayed elastic contributions.

2. Modeling procedure

For this study we simulate eleven different alkali aluminosilicate compositions according to the formula $x\text{Na}_2\text{O}$ ($20-x$) K_2O $20\text{Al}_2\text{O}_3$ 60SiO_2 (mol%), where $x=0, 2, 4, \dots, 20$. The glass compositions are designed to cover the full range of alkali concentrations from pure Na_2O to pure K_2O end members. The procedure for preparing the simulated glass structures is similar to that of our previous study of alkali tetrasilicate glasses published in Ref. [11]. We begin by placing ~20,000 atoms at random positions in a cubic simulation cell. The initial dimensions of the cell are chosen to match experimental data in literature [19]. Periodic boundary conditions are applied at all faces of the simulation cell.

The “as-melted” glass structures are prepared via a stepwise quench from high temperature (4200 K), combining Metropolis Monte Carlo (MC) [20] and molecular dynamics (MD) runs at each

step. The temperature is lowered in steps of 300 K from 4200 K to 900 K. A final drop of 200 K brings the glass to its final temperature of 700 K, which corresponds to a typical laboratory temperature for salt bath treatments. At each temperature step in the simulation, we calculate 100 million trial moves using MC under isobaric conditions (i.e., NPT ensemble at atmospheric pressure). The MC runs are conducted using an in-house code, Molym [21–23], with a target acceptance ratio of 0.35 for both atomic displacements and cell volume changes. Equilibration during the MC runs is monitored based on relaxation of the total potential energy and the volume of the simulation cell.

The output structures from the MC runs are used as input for the MD simulations, which are conducted using GROMACS [24]. Each MD run is performed under constant volume (i.e., NVT) conditions for 600 ps using a time step of 1 fs. The output from the MD simulations is used as input for the subsequent round of simulations at the next temperature. At the final temperature of 700 K, we run another MD simulation in the NPT ensemble for 8 ns to calculate the final glass structure and molar volume of the system. Such combination of MC and MD runs is an efficient and accurate procedure for preparing glass structures through atomistic level simulation [25, 26]. All of the simulations are performed using the pairwise interatomic potentials for silicate systems developed by Pedone et al. [27] Electrostatic interactions are calculated using the Wolf method [28] in the MC simulations and using particle mesh Ewald [29] in the MD program.

After the “as-melted” glasses are equilibrated at 700 K, the “ion-exchanged” glasses are prepared by a direct substitution of Na with K atoms. In each glass we consider a complete exchange of all Na atoms with K. This substitution is followed by an MD simulation for 9 ns in the NVT ensemble and then another 5 ns of equilibration in the NPT ensemble. The temperature is held at 700 K (the salt bath temperature) during this equilibration process. Calculation of the LNDC is based on comparing the final molar volume of the system before and after the Na exchange with K. Calculated values of molar volume are accurate within about $\pm 1\%$.

Finally, the “reverse-exchanged” glasses are prepared by replacing the invading potassium atoms in the ion-exchanged glasses with sodium, again at 700 K. Please note that we replace only those potassium atoms that were introduced during the ion exchange process: the potassium atoms present in the original glass structure are *not* exchanged for sodium. After the invading potassium atoms are swapped for sodium, the same relaxation procedure is employed as in the original ion exchange (9 ns of relaxation in the NVT ensemble followed by 5 ns of equilibration in the NPT ensemble). The final molar volume of the reverse-exchanged glasses is then compared to the molar volume of the original as-melted glasses to determine the degree of elasticity versus plasticity in the ion stuffing process.

3. Results

The final molar volumes of the as-melted, ion-exchanged, and reverse-exchanged glasses are plotted in Fig. 1 as a function of sodium concentration in the as-melted glass. As in our previous study of alkali tetrasilicate glasses [11], the as-melted glasses display a linear scaling of molar volume with concentration. This yields a constant value of the linear network dilation coefficient, as calculated from Eq. (2) and indicated by the dashed line in Fig. 2. This represents an “ideal” value of the LNDC in the absence of the network dilation anomaly. Since the ion-exchanged glasses all have the same chemical composition as the as-melted potassium end member, one might expect that they should also have identical values of molar volume. However, Fig. 1 shows that the actual molar volume of the ion-exchanged glasses is significantly less than that of the potassium end member. Moreover, the final molar volume is a strong function of the initial composition of the as-melted glass. This leads to a composition-dependent LNDC for the ion-exchanged glasses. As

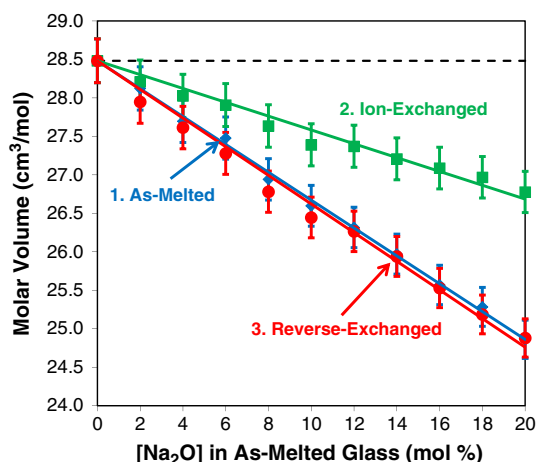


Fig. 1. Composition dependence of molar volume for the as-melted, ion-exchanged, and reverse-exchanged glasses in the $x\text{Na}_2\text{O} (20-x)\text{K}_2\text{O} 20\text{Al}_2\text{O}_3 60\text{SiO}_2$ (mol%) system at 700 K. After the reverse ion exchange process, the as-melted molar volume is recovered for all compositions.

shown in Fig. 2, the value of the LNDC varies from 23% to 55% of the ideal value calculated from Eq. (2). The existence of the network dilation anomaly in these alkali aluminosilicate glasses is consistent with our previous results on the alkali tetrasilicate system [11].

Fig. 1 also shows that the molar volume of the as-melted glasses is completely recovered in the reverse-exchanged glasses, i.e., after the invading potassium atoms are replaced with sodium and again allowed to relax following the same procedure as during the first ion exchange process. This result demonstrates that the strain due to ion stuffing is completely elastic in nature. In other words, network plasticity plays no role in governing the network dilation anomaly. Our simulations indicate that the structure of the as-melted glass remains accessible from the ion-exchange glass, so upon reverse exchange the original as-melted glass structure can be completely recovered. This conclusion is further supported by Fig. 3, where we plot the silicon–alkali pair distribution functions of the as-melted, ion-exchanged, and reverse-exchanged versions of the $[\text{Na}_2\text{O}] = 18 \text{ mol}\%$ glass. The Si-K distribution function shifts to shorter distances in the ion-exchanged glass compared with that of the as-melted glass, taking on intermediate values between the Si-Na and Si-K distributions in the as-melted glass. After the reverse exchange

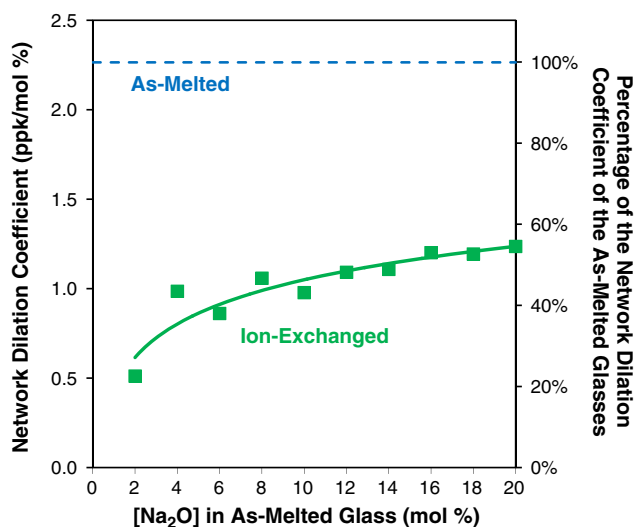


Fig. 2. Linear network dilation coefficient (LNDC) of the as-melted versus ion-exchanged glasses. The LNDC of the ion-exchanged glasses is dependent on composition and ranges from 23% to 55% of the ideal value calculated from the as-melted glasses using Eq. (2).

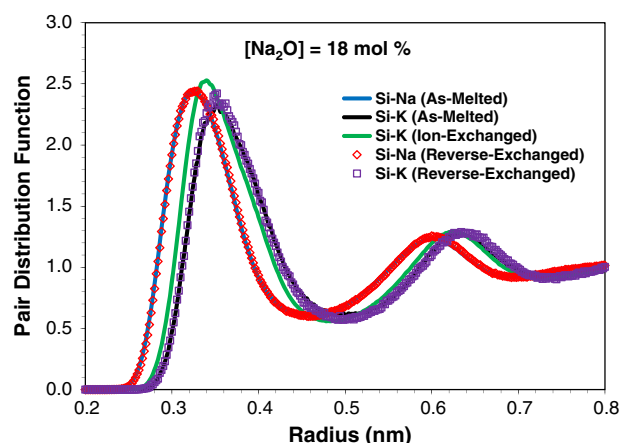


Fig. 3. Calculated silicon–alkali pair distribution functions for the as-melted, ion-exchanged, and reverse-exchanged versions of $18\text{Na}_2\text{O} 2\text{K}_2\text{O} 20\text{Al}_2\text{O}_3 60\text{SiO}_2$ (mol%). The Si-K pair distribution function is shifted to smaller distances in the ion-exchanged version of the glass compared to the original as-melted glass. The pair distribution functions of the as-melted glass are recovered after completing the reverse-exchange process.

process, both the original Si-Na and Si-K distributions are fully recovered. This effect is also shown in Fig. 4 for the aluminum–alkali pair distribution functions. Here the Al-K first neighbor peak again takes on intermediate values between the Al-Na and Al-K distributions of the as-melted glass. As with the silicon–alkali distributions, the Al-Na and Al-K distributions of the reverse-exchanged glass completely overlap those of the original as-melted glass.

This structural recovery of the reverse-exchanged glasses is also clear from the coordination number distributions plotted in Fig. 5. Here the average K-O coordination number of the ion-exchanged glasses adopts an intermediate value between the average K-O and Na-O coordination numbers of the as-melted glasses. When these K atoms are replaced with Na during the reverse exchange process, the original oxygen coordination around the Na sites is again recovered. These results are consistent with our pair distribution functions in Figs. 3 and 4.

While there are significant changes in the local structural environment around potassium between the as-melted and ion-exchanged glasses, Fig. 6 shows that there is no difference in the Q^n speciation of either silica or alumina in these glasses. Here the integer n in the Q^n notation indicates the number of bridging oxygen atoms per silica or alumina tetrahedron. Fig. 6 shows that the distribution of

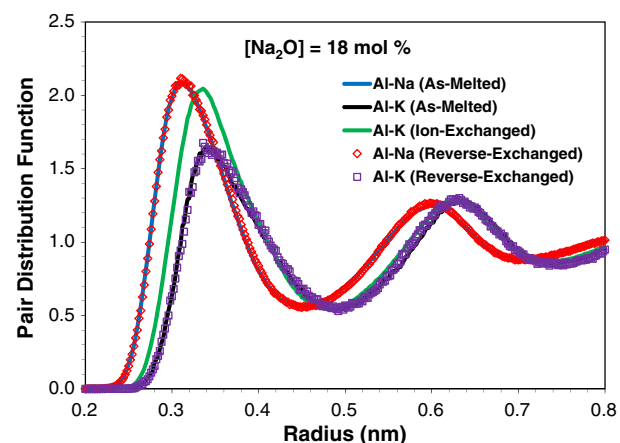


Fig. 4. Calculated aluminum–alkali pair distribution functions for the as-melted, ion-exchanged, and reverse-exchanged versions of $18\text{Na}_2\text{O} 2\text{K}_2\text{O} 20\text{Al}_2\text{O}_3 60\text{SiO}_2$ (mol%). The Al-Na and Al-K distributions of the reverse-exchanged glass overlap those of the original as-melted glass.

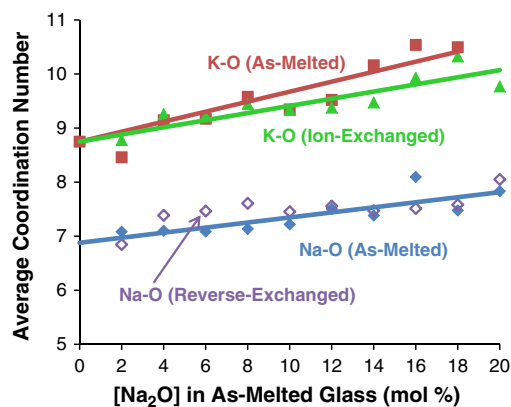


Fig. 5. Average coordination number of oxygen around sodium and potassium. The average coordination of oxygen around potassium is lower in the ion-exchanged glasses compared to the original as-melted glasses. The original oxygen coordination around sodium is recovered after the reverse-exchange process.

nonbridging oxygens is independent of the starting composition of the glass. The Q^n distribution is also found to be independent of whether or not the glasses are ion-exchanged. The final reverse exchange process also has no effect on the Q^n speciation of either Si or Al. Thus, changes in the local environments around the network former cations do not play a role in the mechanical response of the glass to ion stuffing.

4. Discussion

Our current molecular dynamic simulations for alkali aluminosilicate glasses confirm our previous findings from alkali silicate glasses

[11] regarding the atomistic origin of the network dilation anomaly. In both systems, the network dilation anomaly is the result of differences in the local alkali environments between as-melted and ion-exchanged glasses, where the invading potassium ions in the ion-exchanged glasses are unable to achieve the desired structural environment corresponding to that of as-melted versions of the same glass. Despite the differences in the structural role of alkali ions in silicate versus aluminosilicate glasses [30], this explanation of the network dilation anomaly appears to be universal. When the potassium ions are substituted for sodium during the ion exchange process, the average K-O coordination number takes on an intermediate value between what would be expected for potassium in the initial sodium-containing glass and what is ultimately desired in the pure potassium end member. The cation-potassium pair distribution functions also take on intermediate values, consistent with these changes in the oxygen coordination sphere around potassium.

In prior literature [8,13–17], the network dilation anomaly is typically described as a result of plasticity in the glass network, i.e., an irrecoverable deformation of the glass network on a laboratory time scale. For example, the cation-induced relaxation of the network (CIRON) model proposed by Ingram et al. [17] involves an evolution of the Q^n speciation of silica leading to stress relaxation. The presence of such network plasticity would allow for relaxation of the stress due to ion stuffing. Thus when Eq. (1) is applied to determine the LNDC, the resulting value would be lower than that obtained from Eq. (2). However, the results of our reverse exchange simulations clearly show that the strain induced by ion stuffing is entirely elastic in nature. Moreover, the proposed changes in Q^n speciation as predicted by the CIRON model are not observed in our simulations (see Fig. 6).

The network dilation anomaly is therefore not attributable to network plasticity, but rather to a failure of the glass to achieve the fully expected elastic strain due to structural limitations. Plasticity does, however, play a role in chemically strengthened glass as it pertains to stress relaxation. For example, if an ion-exchanged glass is held at an elevated temperature for a sufficiently long period of time, further rearrangements of the glass structure will enable relaxation of the stresses installed during the ion exchange process. Expressed in terms of strain, stress relaxation entails a conversion of the recoverable elastic strain due to ion stuffing to an irrecoverable plastic strain. The physics of stress relaxation and the conversion of

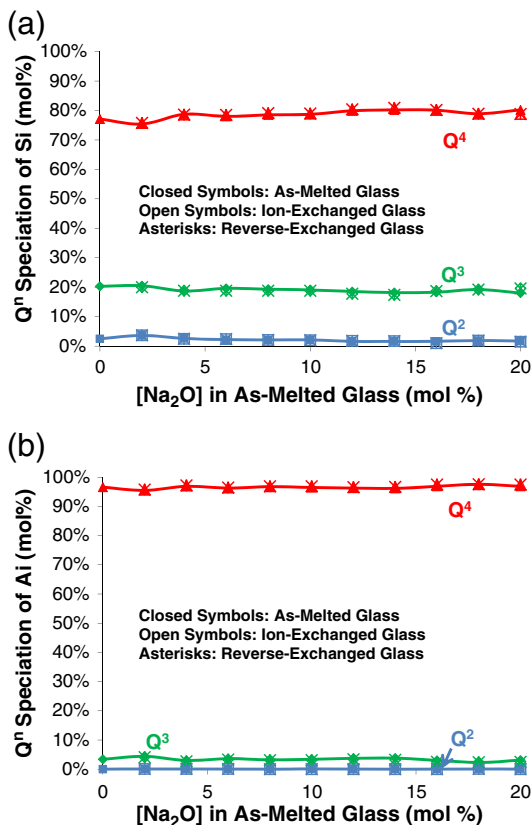


Fig. 6. Q^n speciation of (a) silica and (b) alumina in the as-melted, ion-exchanged, and reverse-exchanged glasses. There is no discernible difference in the Q^n speciation due to the ion-exchange or reverse-exchange processes.

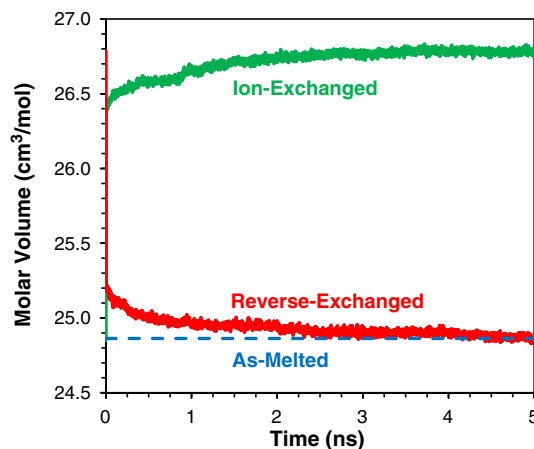


Fig. 7. Time dependence of molar volume during the ion exchange and reverse exchange processes for 20Na₂O 20Al₂O₃ 60SiO₂ (mol%). The final molar volume of the reverse-exchanged glass is identical to that of the original as-melted system, indicating that the strain installed through ion stuffing is completely elastic in nature. For both the K⁺-for-Na⁺ and reverse Na⁺-for-K⁺ exchanges, the elastic response consists of instantaneous and delayed elastic components. For this glass, the instantaneous response accounts for about 85% of the total elastic strain, and the delayed elastic response accounts for the remaining 15%.

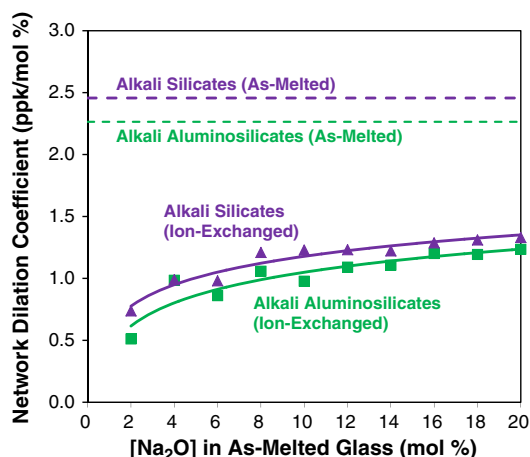


Fig. 8. Calculated linear network dilation coefficient of $x\text{Na}_2\text{O} (20-x)\text{K}_2\text{O} 80\text{SiO}_2$ and $x\text{Na}_2\text{O} (20-x)\text{K}_2\text{O} 20\text{Al}_2\text{O}_3 60\text{SiO}_2$ (mol%) glasses. The “as-melted” lines are calculated from Eq. (2) using the molar volumes of the as-melted glasses. The “ion-exchanged” curves are computed through direct substitution of Na^+ for K^+ in the molecular dynamics simulations.

elastic to plastic strain is the subject of a thorough experimental and theoretical investigation by Potuzak et al. [31]

A more detailed look at our simulation results reveal that the elastic strain experienced by a glass during ion stuffing consists of both instantaneous and delayed elastic responses. For example, Fig. 7 plots the time dependence of molar volume during the ion exchange and reverse exchange processes for the sodium end member: $20\text{Na}_2\text{O} 20\text{Al}_2\text{O}_3 60\text{SiO}_2$. These curves are obtained through the NPT equilibration phase of our molecular dynamics simulations, as described in Sect. II. During the first K^+ -for- Na^+ ion exchange process, the molar volume experiences a near-instantaneous jump followed by a slower delayed elastic response to the final molar volume. The instantaneous response accounts for about 85% of the total elastic strain due to ion exchange, and the delayed elastic response accounts for the remaining 15%. After undergoing a reverse (Na^+ -for- K^+) exchange, the final molar volume of the glass is equivalent to that of the original as-melted glass, demonstrating the pure elastic nature of the ion stuffing process. Again, there is an initial instantaneous response accounting for ~85% of the elastic recovery, followed by a delayed elastic response on the same time scale as during the first ion

exchange process. This demonstrates symmetry in the elastic recovery, with separate instantaneous and delayed ion exchange components that mirror each other during the forward and reverse ion exchange processes.

Finally, it is interesting to compare the results from our current simulations of alkali aluminosilicate glasses ($x\text{Na}_2\text{O} (20-x)\text{K}_2\text{O} 20\text{Al}_2\text{O}_3 60\text{SiO}_2$) with those from our previous study [11] of the alkali tetrasilicate system ($x\text{Na}_2\text{O} (20-x)\text{K}_2\text{O} 80\text{SiO}_2$), since aluminosilicate glasses are known to provide greatly enhanced alkali mobility compared to pure silicate systems [32]. Fig. 8 shows a comparison of the LNDC values of these two systems. We find that the aluminosilicate glasses exhibit a slightly lower LNDC compared to the tetrasilicate systems, indicating that the aluminosilicate glasses would experience less strain for the same degree of ion exchange. However, the higher values of Young's modulus for the aluminosilicate glasses are expected to convert this strain into a greater stress, as indicated by Eq. (1). Fig. 9 plots the same LNDC data as a percentage of the ideal LNDC calculated from Eq. (2) using the molar volume of the as-melted glasses. Interestingly, when normalized to this ideal value, the LNDC values of the aluminosilicate and tetrasilicate glasses collapse onto a single master curve. This demonstrates the apparent universality of the physics governing the network dilation anomaly, despite differences in the local structural environment between alkali silicate and alkali aluminosilicate glasses.

5. Conclusions

The network dilation anomaly imposes a limit to the amount of compressive stress that can be installed in a given glass during ion exchange. This anomaly is apparently a universal phenomenon that results from the different local environments preferred by larger versus smaller ions. In contrast to previous speculations, the network dilation anomaly is not a result of plasticity in the glass network. Rather, it is fundamentally a failure to achieve the full amount of elastic strain expected from as-melted glasses.

Through molecular dynamics simulations, we have shown that the strain induced during ion stuffing is elastic in nature. In other words, the strain that is induced during ion exchange is completely recoverable after performing a reverse ion exchange to restore the original alkali species in the glass. Thus the deformations during ion exchange may be appropriately classified as nonlinear elasticity. Our simulations also reveal that the elastic response of the glass to ion stuffing consists of both instantaneous and delayed elastic contributions. The elastic nature of the strain due to ion stuffing is key in achieving the desired compressive stress in an ion-exchanged glass.

Acknowledgments

We are grateful for the stimulating discussions with Arun K. Varshneya of Alfred University and with Douglas C. Allan, Roger J. Araujo, Phong Diep, Adam J. Ellison, Amy L. Rovelstad, and Natesan Venkataraman of Corning Incorporated.

References

- [1] L. Wondraczek, J.C. Mauro, J. Eckert, U. Kühn, J. Horbach, J. Deubener, T. Rouxel, *Adv. Mater.* 23 (2011) 4578.
- [2] S.S. Kistler, *J. Am. Ceram. Soc.* 45 (1962) 59.
- [3] R. Gy, *Mater. Sci. Eng. B* 149 (2008) 159.
- [4] A.K. Varshneya, *Int. J. Appl. Glass Sci.* 1 (2010) 131.
- [5] A.K. Varshneya, *J. Non-Cryst. Solids* 356 (2010) 2289.
- [6] S. Karlsson, B. Jonson, C. Ståhlhanske, *Glass Technol. Eur. J. Glass Sci. Technol. A* 51 (2010) 41.
- [7] A.R. Cooper, D.A. Krohn, *J. Am. Ceram. Soc.* 52 (1969) 665.
- [8] H.M. Garfinkel, C.B. King, *J. Am. Ceram. Soc.* 53 (1970) 686.
- [9] J. Shen, D.J. Green, *J. Non-Cryst. Solids* 344 (2004) 79.
- [10] V. Tyagi, A.K. Varshneya, *J. Non-Cryst. Solids* 238 (1998) 186.
- [11] A. Tandia, K.D. Vargheese, J.C. Mauro, A.K. Varshneya, *J. Non-Cryst. Solids* 358 (2012) 316.
- [12] J.C. Mauro, R.J. Loucks, *J. Non-Cryst. Solids* 355 (2009) 676.
- [13] D.K. Hale, *Nature* 217 (1968) 1115.

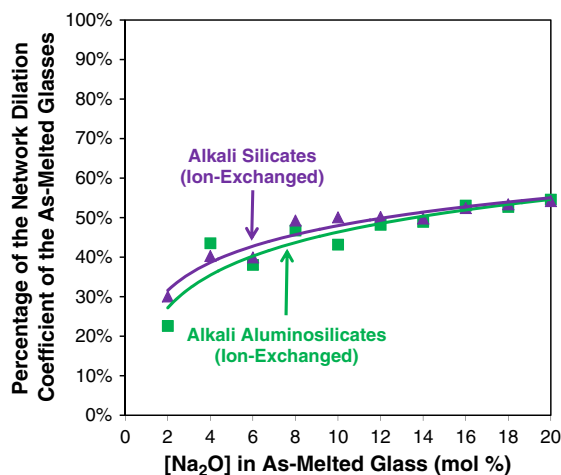


Fig. 9. The network dilation coefficients of the ion-exchanged glasses in Fig. 8 plotted as a percentage of the network dilation coefficient of the corresponding as-melted glasses. When plotted in this manner, the alkali silicate and alkali aluminosilicate glasses collapse onto a single master curve.

- [14] A.K. Varshneya, J. Am. Ceram. Soc. 58 (1975) 106.
- [15] V. Jain, A.K. Varshneya, J. Am. Ceram. Soc. 70 (1987) 595.
- [16] L.B. Glebov, V.G. Dokuchaev, N.B. Nikonorov, G.T. Petrovskii, Sov. J. Glass Phys. Chem. 14 (1989) 131.
- [17] M.D. Ingram, J.E. Davidson, A.M. Coats, E.I. Kamitsos, J. Kapoutsis, Glastech. Ber. Glass Sci. Technol. 73 (2000) 89.
- [18] Y.B. Fu, R.W. Ogden (Eds.), Nonlinear Elasticity, Cambridge University Press, Cambridge, 2001.
- [19] O.V. Mazurin, J. Non-Cryst. Solids 351 (2005) 1103.
- [20] N. Metropolis, A.W. Rosenbluth, M.N. Rosenbluth, A.H. Teller, E. Teller, J. Chem. Phys. 21 (1953) 1087.
- [21] J.C. Mauro, A.K. Varshneya, J. Am. Ceram. Soc. 89 (2006) 2323.
- [22] J.C. Mauro, A.K. Varshneya, J. Non-Cryst. Solids 353 (2007) 1226.
- [23] J.C. Mauro, A.K. Varshneya, J. Am. Ceram. Soc. 90 (2007) 192.
- [24] H.J.C. Berendsen, D. Van der Spoel, R. Van Drunen, Comput. Phys. Commun. 91 (1995) 43.
- [25] K.D. Vargheese, A. Tandia, J.C. Mauro, J. Chem. Phys. 132 (2010) 194501.
- [26] A. Tandia, N.T. Timofeev, J.C. Mauro, K.D. Vargheese, J. Non-Cryst. Solids 357 (2011) 1780.
- [27] A. Pedone, G. Malavasi, M.C. Menziani, A.N. Cormack, U. Segre, J. Phys. Chem. B 110 (2006) 11780.
- [28] D. Wolf, P. Keblinski, S.R. Phillpot, J. Eggebrecht, J. Chem. Phys. 110 (1999) 8254.
- [29] T. Darden, D. York, L. Pedersen, J. Chem. Phys. 98 (1993) 10089.
- [30] Q.J. Zheng, M. Potuzak, J.C. Mauro, M.M. Smedskjaer, R.E. Youngman, Y.Z. Yue, J. Non-Cryst. Solids 358 (2012) 993.
- [31] M. Potuzak, R.C. Welch, J.C. Mauro, J. Chem. Phys. 135 (2011) 214502.
- [32] M.M. Smedskjaer, Q.J. Zheng, J.C. Mauro, M. Potuzak, S. Mørup, Y.Z. Yue, J. Non-Cryst. Solids 357 (2011) 3744.